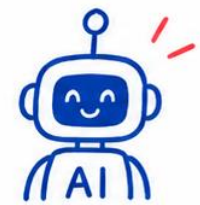
Computer Science Education in the Age of AI

Policy Directions and Challenges

KEY QUESTION



What should **every student** learn when **intelligence itself** becomes part of the computing infrastructure?



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PART I

Korea's Computing Education Journey



Korea's K-12 Computing Curriculum Journey

From ICT use to computational thinking, AI and data—and the generative AI challenge

1. 2000s

ICT Utilization Education



- Using digital tools in learning
- Digital access and use

2. 2015 Revised Curriculum

Compulsory middle-school Informatics



- Computational thinking
- Programming and software

3. 2022 Revised Curriculum

Digital literacy, data, and AI

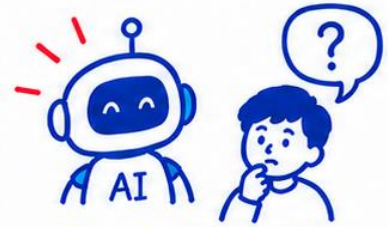


- Foundational digital literacy
- Data and AI basics

4. 2023+

New challenge!

Generative AI challenge



- Foundation models and prompts
- Rethinking curriculum goals

Korea has moved from using technology, to learning computing, and now to rethinking education in an AI-powered society.

PART Ⅱ

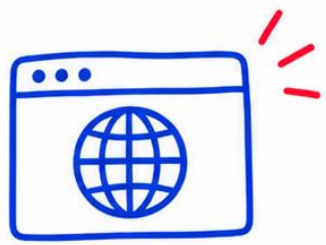
Computing Infrastructure and Educational Change

From web to mobile to AI



Three Shifts in Computing Infrastructure

An interpretive framework: new layers accumulate rather than replace earlier ones.



1990s–2000s

Internet + Web

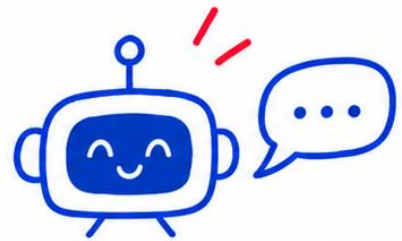
- Networked information and web services
- Digital literacy and ICT use



2010s

Smartphone + Apps

- Ubiquitous access and platform ecosystems
- Computational thinking and software creation



Now

AI Foundation Models + Agents

- Reusable AI capabilities across applications
- AI literacy, stronger CS foundations, and human–AI problem solving

New layers accumulate; they do not replace what came before.

Sources: CERN; Apple (2018); Bommasani et al. (2021); Wang et al. (2024).

AI as an Emerging Computing Infrastructure

Foundation models, cloud, and data are becoming a basic infrastructure for society.

AI is not just an app — it is becoming a basic infrastructure.

3 Society Re-designed

AI is reshaping every part of society and our lives.

Education | Economy | Industry | Culture | Everyday Life

Enables new services and transforms society

2 Software and AI Agents

Software and agents build on the infrastructure layer.

Applications and Services

AI Agents (Assistants, Copilots, Automation)

APIs and Developer Tools

Built on top of cloud, data, compute, and foundation models

1 Basic Infrastructure

The reusable infrastructure that makes AI possible.

Cloud (scalable and reliable)

Data (storage and pipelines)

Compute (GPUs and specialized hardware)

Foundation Models (general-purpose intelligence)

AI is becoming a **foundational computing infrastructure** that powers software, agents, and the redesign of society.

Technology Reshapes Society and Education

Capabilities accumulate rather than replace one another.

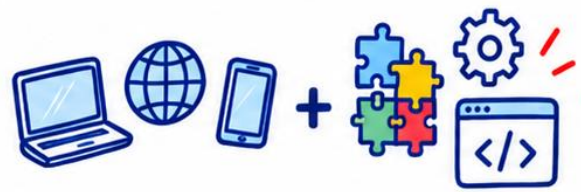
Each new capability builds on what came before.

1 Digital Literacy



- access and use digital tools
- participate in digital environments

2 Digital Literacy + Computational Thinking



- solve problems with computation
- create with software and data

3 Digital Literacy + Computational Thinking + AI Literacy



- understand and use AI
- evaluate AI outputs and limitations

New capabilities extend—not replace—earlier foundations.

From Programming to AI-Enabled Computing

How the role of programming is changing

1 Earlier

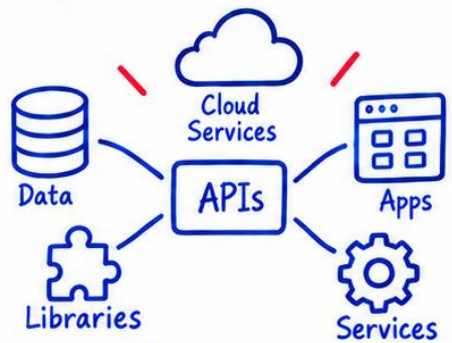
Write



- algorithms
- code
- debugging

2 Platform era

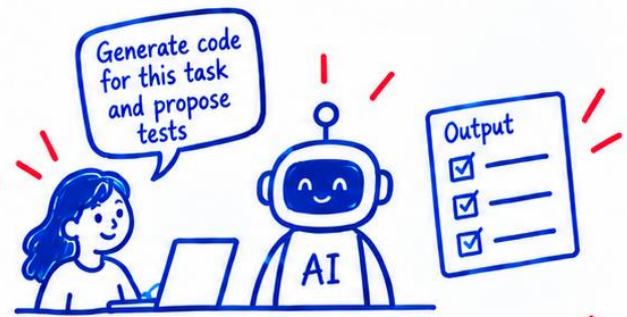
Compose and Integrate



- libraries
- APIs
- frameworks

3 AI era

Specify, Delegate, and Verify



- describe goals
- use AI tools
- test and integrate

Human responsibility remains essential.

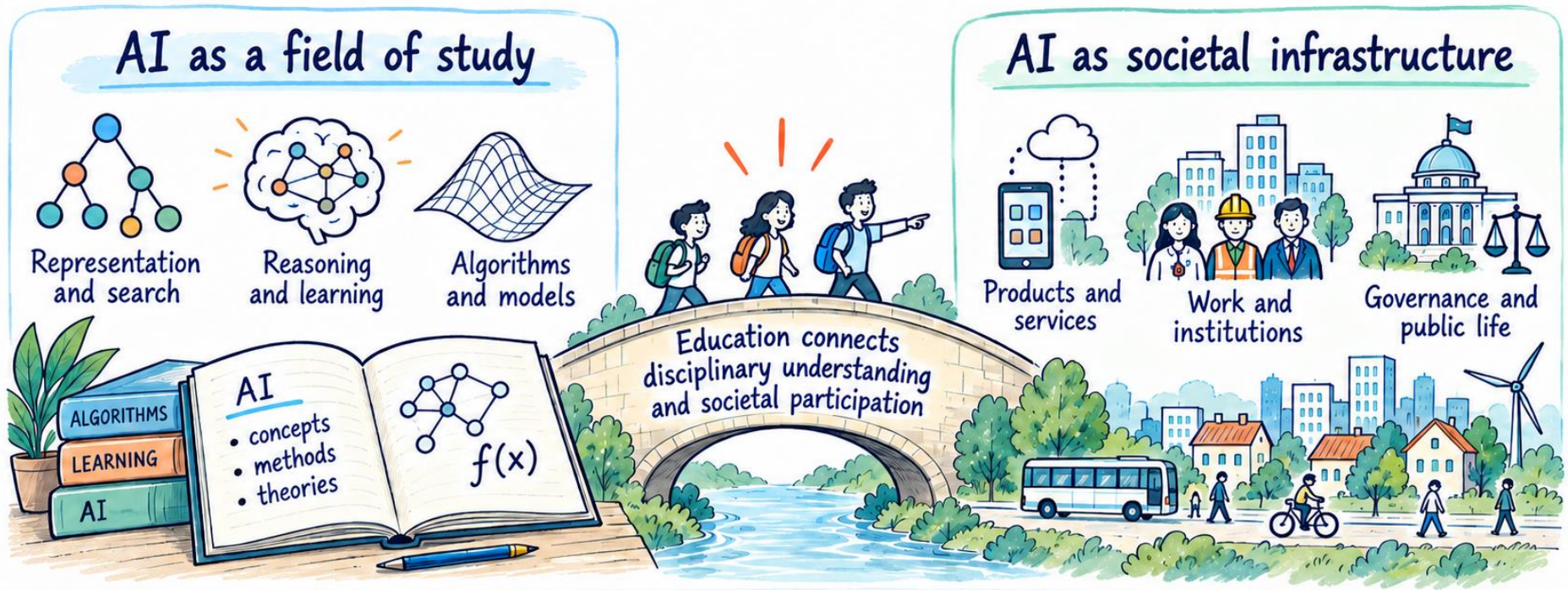
From writing every line by hand to composing, orchestrating, and verifying computation with AI.

PART III

Rethinking CS and AI Education



AI as a Discipline and as Infrastructure

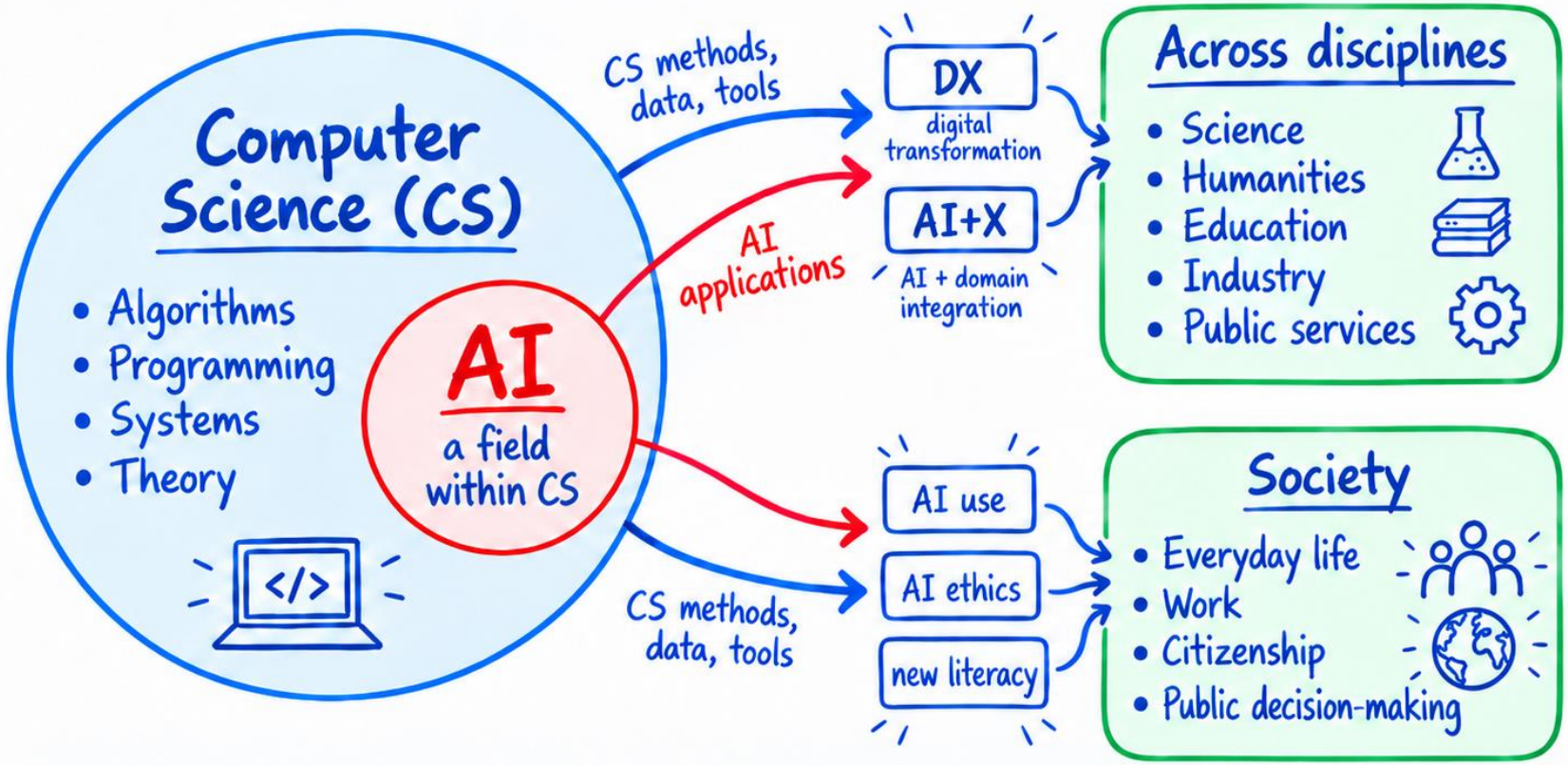


Students need to learn AI both as knowledge to understand and as infrastructure to live and act with.

Author's synthesis

CS Foundations for AI and Society

CS and AI connect to disciplines, society, and new forms of literacy.



We need strong CS foundations, meaningful AI applications, and AI literacy for society.

What Must We Preserve and Extend?

Building on the strengths of CS, preparing for an AI-driven future

What to Preserve
(Core of CS Education)

Algorithms & Problem Solving
- algorithmic thinking
- efficiency and correctness
- classic problem solving

Programming
- expressing ideas
- building and debugging
- creative construction

Computational Thinking
- decomposition
- abstraction and modeling
- logical and systematic thinking

Data and Information
- data representation
- data analysis and visualization
- data-driven decision making

Computing Systems
- hardware, software, networks
- operating systems
- how computing systems work

A Strong Foundation for the Future

From CS to CS + AI

What to Extend
(New in the AI Era)

Understanding and Using AI
- AI concepts and how it works
- generative AI and foundation models
- effective and critical use

Human-AI Collaboration
- augmented intelligence
- new ways of working and learning
- creativity with AI

Verification and Evaluation
- judging the reliability of AI outputs
- understanding bias and limitations
- evidence-based evaluation

Human Values and Responsibility
- AI ethics and safety
- privacy, fairness, and human rights
- social and environmental impact





Computational Thinking and Problem Solving

How can computation help us solve a problem?

1 Represent the problem



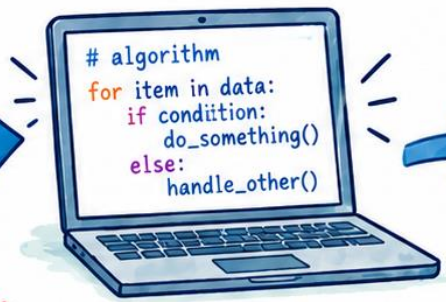
Identify the relevant information, data and context.

2 Abstract and decompose



Focus on what matters, identify key components, and break the problem down.

3 Develop a computational solution



AI can assist here — but learners remain responsible for all four steps.

4 Test, evaluate, and improve



Run tests, check results, find issues, and make it better.

Iterate and improve

Programming makes ideas executable and testable.

Testing and evaluation make them trustworthy.

Human Reasoning, Judgment, and Responsibility

AI can generate. Humans must judge.

AI Outputs



- text
- code
- image
- suggestion

AI can assist, but it does not replace human judgment.

Interpret



Evaluate



Judge



Decide & Take Responsibility



Key questions



- Is it correct?
- What evidence supports it?
- What assumptions or biases are present?
- Is it appropriate in this context?

What learners must do



- understand the context
- check accuracy and evidence
- evaluate limitations and bias
- make a human judgment
- take responsibility for decisions

The more capable AI becomes, the more important human judgment becomes.

A Proposed Curriculum Framework for the AI Era

Literacy for all · Disciplinary depth · Integration across subjects

3 AI and Digital Integration across Subjects

Use and apply AI and digital technologies to explore, create, and solve real-world problems in all subjects



2 Informatics-Centered CS, Software, Data, and AI Education

Learn the core concepts and practices of computer science and AI through the Informatics subject and related courses



1 Foundational Digital and AI Literacy for Every Student

Develop the basic knowledge, skills, and attitudes to understand, use, and live with AI in a safe, critical, and responsible way



Across all K–12 levels, with increasing depth and sophistication



All students, at every level, benefit from all three layers.

All three layers are necessary because AI is both a field of study and a societal infrastructure.

Author's synthesis

PART IV

Learning Pathways from School to University



International Perspectives on Learning, CS, and AI

Different contexts, common directions



UNESCO (2024)

AI Competency Framework for Students

- 12 competencies across 4 dimensions
- Human-centred mindset and ethics of AI
- Progression: Understand → Apply → Create
- Students as responsible users and co-creators



OECD

Learning Compass 2030

- Student agency and well-being
- Knowledge, skills, attitudes, and values
- Transformative competencies
- Learning to act responsibly in a changing world



CSTA (2026)

PK–12 Computer Science Standards

- Computational thinking and practices
- Inclusive and equitable CS education
- Real-world computing and AI connections
- Preparing every student for a world powered by computing



CS2023

Undergraduate Computer Science Curricula

- 17 knowledge areas
- Competency-oriented curriculum design
- Society, ethics, and the profession integrated across CS
- Includes guidance on generative AI in CS education

Common message: prepare all learners with computing knowledge, responsible judgment, ethics, and readiness for an AI-enabled world.

Sources: UNESCO, AI Competency Framework for Students (2024); OECD Learning Compass 2030; CSTA PK–12 Standards (2026); CS2023 Executive Summary (2024).

A Coherent K-12 Learning Pathway

From foundational literacy to advanced application and real-world impact

Progressive, connected, real-world oriented

Growing opportunities for all learners

Elementary School



Explore and Build Foundations

- Basic digital and AI literacy
- Playful computational thinking
- Understanding technology in daily life

Curiosity and confidence

Middle School



Learn and Apply

- Core concepts of CS, data, and AI
- Programming and design activities
- Problem solving and social understanding

Skills and computational problem solving

High School



Deepen and Create

- Advanced CS and AI topics
- Real-world and interdisciplinary projects
- Ethics, society, and career pathways

Creation and real-world connection

Beyond School



Lifelong Learning

- Continuous learning and upskilling
- Adaptation to new technologies
- Participation in an AI-enabled society

Lifelong growth and societal contribution

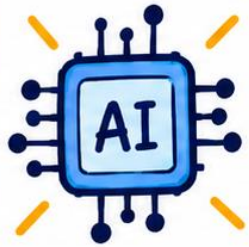
Building knowledge, skills, values, and agency across the entire K-12 journey

Learning pathways should build progressively while keeping all learners connected to computing and AI.

Universities in Transition

Expanding roles for AI in higher education

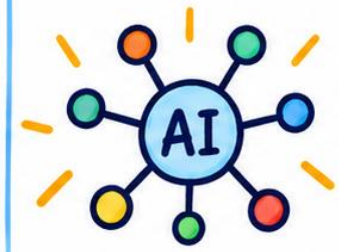
AI as a Field within CS



Deepen and advance AI as a core area of study and research.

- Fundamental theories and methods
- Specialized courses and research
- Next-generation AI talent

AI across the CS Curriculum



Integrate AI throughout the computing curriculum.

- AI as a tool and a topic in many areas
- Hands-on experience and real-world projects
- Preparing all CS students for an AI-enabled world

AI across Disciplines



Collaborate beyond CS for real-world impact.

- AI for major-specific applications
- Interdisciplinary programs and joint courses
- Addressing societal challenges with AI

Universities play a key role in developing **AI talent**, **advancing knowledge**, and fostering **interdisciplinary innovation**.

PART V

Policy and Practice for CS and AI Education



Aligning Curriculum, Teaching, and Assessment

A Coherent System for Meaningful Learning

Aligned
Consistent
Reinforcing

Curriculum

What to Learn?

Standards
- Knowledge
- Skills
- Values

- AI literacy for all
- Core CS concepts
- AI + X (interdisciplinary)
- Ethics, society and global issues

Clear goals, progressive structure, relevant and up-to-date content

Teaching & Learning

How to Learn?



- Student-Centered
- Active Learning
- Project-Based
- Real-World Contexts
- Human-AI Collaboration

Engaging, inclusive and practice-oriented learning experiences

Assessment

What and How Well?

- Measure diverse outcomes (knowledge, skills, attitudes)
- Performance-based assessment
- Projects, portfolios, real-world tasks
- Formative feedback with AI tools
- Support learning and improvement (not just ranking)

Fair, authentic, and future-oriented assessment

Continuous Improvement
(Curriculum ↔ Teaching ↔ Assessment)

Today's Learners



Meaningful Learning for a Better Future



Confident
Creative
Responsible
Global Citizens

A Coherent Policy Ecosystem for AI Education

Lessons from Korea's AI education action plans

From policy directions to an aligned education ecosystem

KOREA
AI EDUCATION
FOR A BRIGHTER
TOMORROW

1. Curriculum Reform



- clear learning goals
- AI, data, and CS integration
- progressive K-12 pathways

2. Teacher Capacity



- teacher learning and support
- pedagogical design with AI
- professional judgment

3. Safe Educational AI



- privacy and security
- reliability and transparency
- age-appropriate use

4. Bridging the AI Divide



- access and equity
- inclusive participation
- digital opportunity

5. Governance & Ecosystem



- partnerships and coordination
- research and infrastructure
- continuous improvement

AI for All

Working together
for an inclusive, ethical,
and empowered
AI future.

The key challenge is not isolated projects, but connecting these five areas into **one coherent system.**

출처: 국가인공지능전략위원회, 『대한민국 AI 행동계획』 보고서, 2026년 2월.

From Policy to Student Learning: One Aligned System

Vertical coherence and horizontal alignment

1. National Vision and Curriculum



- goals and standards
- three-layer curriculum architecture
- governance and funding

2. Enabling Conditions



- teacher capacity
- learning resources and platforms
- safe educational AI
- access and equity

3. Aligned Learning System

Curriculum
what to learn

Teaching
how to learn

Assessment
evidence of learning

Realizea through classroom practice:

- student-centered learning
- projects and real-world problem solving
- use of AI tools
- discussion on ethics and society

4. Learner Outcomes



- understand computing, data, and AI
- solve and create
- evaluate and exercise judgment
- act responsibly

Evidence and feedback for continuous improvement

Policy works when vertical implementation and horizontal alignment meet in the classroom.

Open Challenges for the Next Curriculum

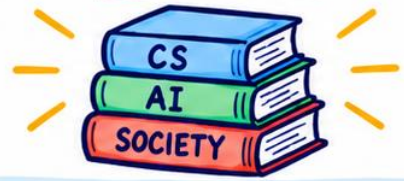
Key questions for the next phase of digital, CS, and AI education

Goals



What should every student learn in the age of AI?

Content



What should be preserved, extended, or reduced?

Teaching and Learning



How should AI change pedagogy and classroom activity?

Assessment



What counts as evidence of real learning when AI can help?

Teachers



What knowledge, support, and judgment do teachers need?

Equity and Governance



How do we ensure access, safety, and public trust?

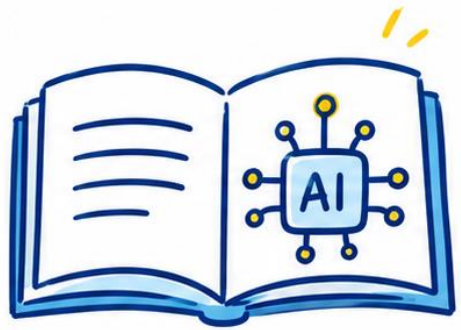
The next curriculum must address not only new technologies, but also new responsibilities.

PART VI

What Should Every Student Learn?



What Should Every Student Learn?



Understand

Understand computing, data, and AI



Create

Solve problems with computing and AI



Judge and Act Responsibly

Evaluate evidence and take responsibility

Stronger CS foundations for every student in the AI era

→ Not less computer science because AI can do more — but broader, stronger CS for every student.

Thank You

Computer Science Education in the Age of AI

Hyeoncheol Kim
Korea University

Questions and Discussion

